TOWARDS WATER-SENSITIVE CITIES

in the age of the Anthropocene



Tanker refill point, Bengaluru, 2024 · Saiphani02, CC BY-SA 4.0

March 2024: too little
6,900 of 14,000 borewells dry; the city turns to tankers

What is a water-sensitive city?

A city that runs **rain, rivers, groundwater and wastewater as one cycle**, and plans its land around where water goes. It holds rain where it falls, recharges the ground and reuses what it drains. Cities tend to move from simply supplying water, to sewering and draining it, to finally working with the whole cycle.

> **Bengaluru's lake cascade once did exactly this. Pipes and drains pushed it backwards: now it both floods and runs dry.**

THREE WAYS TO STUDY IT

Technocratic

Can the system carry the water?

Studies the water itself: how it moves, and what pipes, drains and pumps were built to carry.

Bureaucratic

Which agency is accountable, if any?

Studies the institutions: which agencies own each part of the cycle, and who funds and maintains it.

Political ecology

Who gains, who loses, who is heard?

Studies power: land, money and voice decide where water goes, so floods and shortages fall unevenly.



Tatanagar, Bengaluru, 22 Oct 2024 · Shyamal, CC BY-SA 4.0

October 2024: too much
The same year, Tatanagar floods after heavy rain

Case study: Bengaluru

How political decisions unmade a lake city, 1898–2024

Karnataka, India

DECISION — 1898
Linked lakes (keres) still hold the monsoon. After the plague, planners recast their channels as sanitary drains.

DECISION — 1937–1969
Piped water turns lakes into land: a stadium, then a bus station.

DECISION — 1974
Cauvery water arrives, pumped about 95 km uphill. Lakes look spare.

DECISION — 1990s–2010
The IT boom sells wetlands as real estate. 207 water bodies fall to 93.

TOO MUCH WATER — Sept 2022
131 mm of rain in one night. The Outer Ring Road goes under water.

TOO LITTLE WATER — March 2024
6,900 of 14,000 borewells dry. Tankers sell above the price cap.



Ashwin Kumar, CC BY-SA 2.0

Bellandur Lake, 2018. The city's largest lake, choked with weeds, cleared by machine.

Technical explanation

207 → 93 water bodies, 1973 to 2010 (IISc)

6 of 6 groundwater units over-exploited

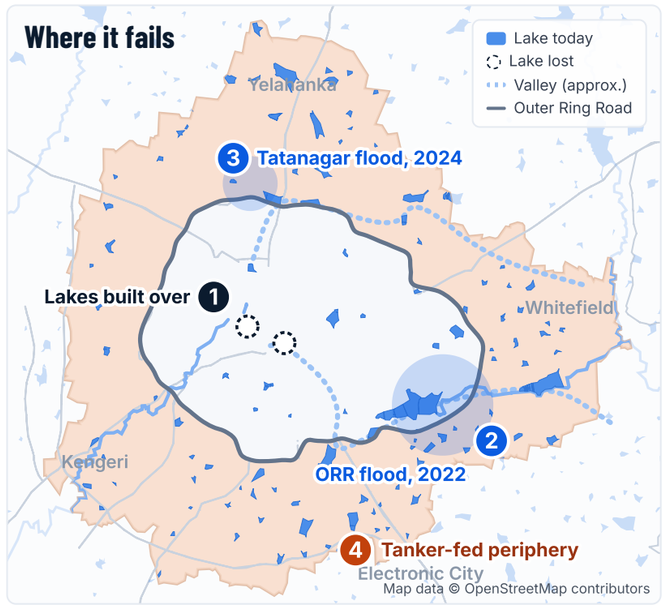
500 / 2,600 MLD short of daily need, 2024

- **Cascade cut:** overflow floods streets instead of filling the next lake.
- **Aquifers:** natural recharge is about a tenth of daily use.
- **Imported:** the city relies on a river about 95 km away.

Political and social explanation

- **Lakes became land:** once piped water arrived, lakes looked like empty land to build on.
- **Selective enforcement:** "encroachment" rules evict settlers on filled channels, but not a multinational on wetland next door.
- **Unequal supply:** only 10–30% of periphery homes had piped water; tanker "mafias" work alongside officials.
- **Rules as a choice:** the state picks when its rules apply. The 2024 tanker price cap stayed on paper.

Where it fails



Solutions for Bengaluru

One answer in each register: the pipes, the offices, and who the water reaches.

Technocratic

- Reconnect lake cascades and protect the valleys they drain.
- Refill lakes with treated wastewater, a large untapped source.
- Build recharge wells so rain soaks back into the ground.

Bureaucratic

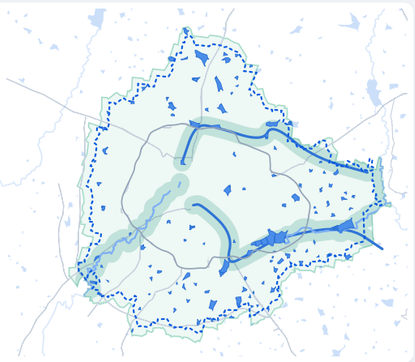
- One catchment authority for lakes, storm drains and sewage.
- Register and meter every tanker, and publish the data.
- Enforce the tanker price cap, not only announce it.

Political ecology

- Apply encroachment rules to corporate and state projects first.
- Secure tenure before any eviction for lake or drain works.
- Give lake users and the periphery a seat in restoration.

Bengaluru, rebuilt

Lakes linked again along protected valleys, and piped water reaching the whole city, not just the core.



Conclusion

Bengaluru once held its rain. Politics, not just hydrology, broke the cascade. A water-sensitive Bengaluru must rebuild the lakes and the rules together.

The right to the city includes the power to decide where water goes and whom it reaches.

Sources: IISc (Ramachandra & Aithal); WELL Labs, 2024; Deccan Herald and The Hindu, March 2024; RCC *Arcadia*. Photos: Wikimedia Commons. Maps © OpenStreetMap contributors; valleys approximate. **Readings:** Wong & Brown 2009; Kaika & Swyngedouw 2000; Roy 2005; Ranganathan 2014, 2015; Hädrich Silva et al. 2024; Harvey 2008.